

Question1: Answer the following:

(4 + 8 M)

1. Given $x = 30.158$ rounded to $\tilde{x} = 30.16$ then :

- Calculate the absolute error between the true value and the computed value.
- Calculate the relative error as a percentage between the true value and the computed value.

a) Absolute error = | Exact value – Approximate value | = | 30.158 – 30.16 | = 0.002

b) Relative error = | Exact value – Approximate value | / | Exact value | =
| 0.002 | / | 30.158 | = $0.66 \times 10^{-4} = 0.0066\%$

2. Given the following polynomial $P(x) = 4x^3 - 8x^2 + 14x - 32$.

- Use Horner's algorithm to evaluate $P(3)$.
- Using the complete Horner's algorithm, find the Taylor expansion of $P(x)$ about the point $r = 2$.

a) 3

$$\begin{array}{r} 4 \quad -8 \quad 14 \quad -32 \\ 3 \quad \quad 12 \quad 12 \quad 78 \\ \hline 4 \quad 4 \quad 26 \quad 46 \end{array}$$

Thus, we obtain $P(3) = 46$.

1

b) 5

$$\begin{array}{r} 4 \quad -8 \quad 14 \quad -32 \\ 2 \quad \quad 8 \quad 0 \quad 28 \\ \hline 4 \quad 0 \quad 14 \quad -4 \\ \quad \quad 8 \quad 16 \\ \hline 4 \quad 8 \quad 30 \\ \quad \quad 8 \\ \hline 4 \quad 16 \end{array}$$

Thus, we can write:

$$P(x) = 4(x - 2)^3 + 16(x - 2)^2 + 30(x - 2) - 4$$

2

Question 2: Floating-point representation**(7 M)**

Determine the double-precision machine representation in a 64-bit word-length computer of the following decimal number (-184.125) .

Solution:

1. Converting the integer part to binary, we have:

$$(184)_{10} = (270)_8 = (1011\ 1000)_2. \quad 1$$

2. Converting the fractional part to binary, we have:

$$(.125)_{10} = (.1)_8 = (.001)_2. \quad 1$$

3. $(1259.125)_{10} = (1011\ 1000.001)_2 = (1.011\ 1000001)_2 \times 2^7$ is the corresponding one-plus form in base 2, and $(.011\ 1000001)_2$ is the stored mantissa. **1**

4. Next the exponent is $(7)_{10}$, then $c - 1023 = 7$, we immediately see that $(130)_{10} = (2006)_8 = (100\ 0000\ 0110)_2$ is the stored exponent. **2**

Thus, the double -precision machine representation of (-184.125) is:

$$[1100\ 0000\ 0110\ 011\ 10000010000000000000]_2 \quad 1$$

$$[1100\ 0000\ 0110\ 0111\ 0000\ 0100\ 0000000000]_2 = [C0670400]_{16} \quad 1$$

Question 3: Given the following function:**(3 + 8 M)**

$$f(x) = 2x^2 - e^x$$

1. In a bisection method if you asked to find a root of an equation in interval $[1, 2]$ with an accurate level of $\varepsilon = 10^{-2}$, how many iterations does it take to find the root?
2. Find a root of the function $f(x)$ on the interval $[0, 2]$ by using the bisection method. Terminate the iterations if they exceed $n_{max} = 2$ or if the error bound is less than 0.5×10^{-6} .

Solution:

1. **3**

$$n > \frac{\log(b - a) - \log 2}{\log 2}$$

$$n > \frac{\log(1) - \log 2 * 10^{-2}}{\log 2} = 5.6 \approx 7$$

- 2.

$$u = f(0) = -1$$

$$v = f(2) = 2 \cdot 2^2 - e^2 = 8 - 7.3891 = 0.6109$$

Since, $u \cdot v < 0$ then the function f has exactly one real root α in the interval $[0, 2]$. **2**

- 2.

- 1) $n = 0$ and $c = (0 + 2)/2 = 1$, $w = f(c) = f(1) = -0.71828$
 $u, w > 0$ then: $a = c$, $b = b$ and the interval $[1, 2]$. Error = 1. 2
- 2) $n = 1$ and $c = (1 + 2)/2 = 1.5$, $w = f(1.5) = 0.01831$
 $u, w < 0$ then: $a = a$, $b = c$ and the interval $[1, 1.5]$. Error = 0.5. 2
- 3) $n = 2$ and $c = (1 + 1.5)/2 = 1.25$, $w = f(1.25) = -0.36534$
 $u, w > 0$ then : $a = c$, $b = b$ and the interval $[1.25, 1.5]$. Error = 0.25. 2

n	cn	$f(cn)$	Error
0	1	-0.71828	1
1	1.5	0.01831	0.5
2	1.25	-0.36534	0.25